OCR NOTES: POSTERS, CAROUSELS, NESTED BOXES

- **Date:** 2026-08-30
- **Status:** idea / architecture note (not a spec, not scheduled work)
- **Scope:** how `extract` should treat **designed images** (social infographics, carousels, two-panel posters) whose argument lives *inside nested boxes*, not in a PDF text layer
- **Related:**
  - 17-local-dev-stack-architecture.md (`extract` profiles: Docling default, jobs not daemons)
  - 18-graph-rag-wikipedia-db-pedia-ieml.md (capability vs Implementation)
  - ASC: Revival v3 §10 (extract profiles, MinerU as metered `layout`)
  - ASC: Revival v5 (extract-once, modest hardware, closable Technologies)
  - ASC: Social media Literature Review (carousel OCR already treated as a *reading* of slides, not an import of the genre)

This note records 2026-08-30 experiments on one representative slide, the tool survey that followed, and a **multi-step extract design**. It does not change engine identity: Docling on CPU remains the default `layout` Implementation for mixed personal files. Posters are a **document class** with their own profile, not a reason to replace Docling with MinerU or a VLM.

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1. THE PROBLEM (WHY `layout` ON A PDF IS THE WRONG PICTURE)

Revival v3 already split `extract` into profiles (`plain`, `layout`, `biblio`, `table`, `figure`). That split assumes a **page**: columns of prose, optional tables, figures that can be cropped and captioned. Social and "AI carousel" images are a different object.
Typical structure (observed on a 2026 educational slide about sliding-window attention / attention sinks):

```
1 column – header paragraph (full width)
2 columns – sibling panels (before / after), each with nested boxes:
            token chips, labels under chips, "MODEL OUTPUT" wells
1 column – footer punchline
```

The argument is **in the diagram**, not in the caption. Nested boxes are not a table and not a figure caption. A layout model trained on papers will often do one of two things:

1. Classify the whole diagram as **one** `picture` and skip inner text in the markdown export.
2. OCR the inner words but **sort left-to-right at the same height**, merging the two panels.

Corpus context (from note 17, anonymized): a social-media export on the order of **~25 GB** / **~28k files**, mostly photos. OCR on every thumbnail is the same cost cliff as ASR on every video. The social-media literature review already refused to OCR all 2 597 saved slide images; it gated on educational infographics. This note is about **that gated class**, not the 17k ordinary photos.
Hardware this house assumes: Debian laptop, i7-8750H, 32 GB RAM, **GTX 1050 Mobile 4 GB** (**Pascal**). Indexes are the intelligence. GPU VLMs ≥1B belong on a later Tiiny (stage 2), not as stage-1 identity.

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2. WHAT WE TRIED (SAME FILE, 2026-08-30)

Source: one JPEG infographic (two outlined panels, nested token boxes, footer "one token evicted."). Tools already on the laptop: `docling` (pipx), RapidOCR / PP-OCRv6. MinerU was **not** installed. Granite-Docling ran as Docling's `--pipeline vlm` (preset `granite_docling`, default).
Working directory for artefacts: `/tmp/docling-infographic*`, `/tmp/granite-docling-infographic`. Re-runnable from the terminal (CLI = GUI).

2.1 Docling standard pipeline (`--pipeline standard`, CPU)

Markdown export (~83 s first run, models already cached afterwards):

- Header paragraph: **correct** (`never earned`, en-dash preserved).
- Two panels: **dropped** as `<!-- image -->`.
- Footer: scrambled to `token evicted. one`.

So the failure is not "Docling mixed the two columns in markdown." Markdown never saw the columns. The layout net labelled the middle as a picture and the exporter did not dump picture children.

2.2 Docling `--ocr-mode full_page`

Markdown: header still correct; footer becomes `one token evicted.` (good); middle still `<!-- image -->`.
JSON (`DoclingDocument`): the inner strings **are** there, as children of `#/pictures/0` (a **single** picture). Reading order is the horizontal merge we feared:

1. `CONTEXT WINDOW` then `CONTEXT WINDOW` again (both panel headers).
2. Token chips from both panels in one stream (`The`, `cat`, `sat`, … then the right panel's chips).
3. `(evicted)`, `PINNED`, `(TOKEN #1)` after both chip rows.
4. `MODEL OUTPUT` then `MODEL OUTPUT`.
5. `fluent.` interleaved with the red-panel garbage (`xq7!@# m9$% zt&*q12^`).
6. Left fluent sentence split across those lines (`The cat sat on the mat and` / `looked out the window.`).

OCR engine here is RapidOCR wrapping **PP-OCRv6** (same family MinerU's pipeline uses for glyphs). Character quality on the gibberish panel is fine enough to copy the slide's fake tokens. **Layout clustering** is what fails.

2.3 Granite-Docling (`--pipeline vlm --vlm-model granite_docling --device cpu`)

~258M VLM, already a Docling preset. ~101 s on CPU for this JPEG. DocTags:

```
<text>…corrupting them with weight they never collapsed. The model doesn't degrade. It collapses.</text>
<picture>…left panel…<other></picture>
<picture>…right panel…<other></picture>
```

JSON confirms two pictures with left edges ~48 vs ~414 (sibling panels). That is the **region split** standard Docling did not do.
Costs:

Piece	Result
Header	Hallucination: `never earned` → `never collapsed` (contaminated by the last sentence).
Two panels	Empty pictures (`<other>`). **No inner OCR.**
Footer	Missing.

Markdown is the damaged paragraph plus two `<!-- image -->` / `Other` placeholders.
Verdict: Granite-Docling is a **region detector** on this class of file, not a nested-box reader. It is useful as a *split*, not as the extract.

2.4 Comparison (same JPEG)

Implementation	Header	Two panels as regions	Inner nested text	Footer	Reading order of panels
Docling standard, md	Good	One picture	Lost in md	Scrambled	n/a
Docling `full_page`, md	Good	One picture	Lost in md	Good	n/a
Docling `full_page`, JSON	Good	One picture	Present	Good	**Merged left↔right**
Granite-Docling VLM	Hallucinated	**Two pictures**	Absent	Lost	Split, empty

No single one-shot tool both split the panels *and* read the nested labels.

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3. MINERU IS THE WRONG FIRST COMPLEMENT (FOR THIS CLASS)

Revival v3 already placed MinerU as a **metered** `layout` **Implementation** for hard *scholarly* pages (tables, formulas, CJK), not as identity, not as a crawl of 4k PDFs.
MinerU's published lead is OmniDocBench (document pages). Pipeline OCR is PP-OCRv6 — same glyph family as Docling's RapidOCR. The accurate path is a VLM that wants **Volta+ GPU and ~8 GB**; this laptop is Pascal 4 GB. CPU `pipeline` can exist as a slow experiment.
For **ordinary social photos**, MinerU is the wrong job (scene text vs document parse). For **this poster**, MinerU *might* segment columns better than Docling's single picture — that is an eval, not a guarantee. Nested infographic chrome (chips, pins, twin "MODEL OUTPUT" wells) is not what OmniDocBench measures. Standing up MinerU before a gold dozen of *these* slides would freeze a heavy Technology on a hypothesis.
Never run Docling + MinerU + Granite + Marker on the same file "to be safe." One path per class, then Fallback.

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4. OPEN-SOURCE TOOLS THAT ACTUALLY TARGET NESTED / MULTI-REGION LAYOUT

None is a "carousel nested-box product." Two families exist.

4.1 Hierarchical layout (detect coarse regions, then elements)

PaddleOCR PP-StructureV3 is the one that names the bug. Two stages:

1. **PP-DocBlockLayout** — large structural regions (newspaper/magazine articles, multi-column blocks).
2. Fine layout — titles, paragraphs, figures, tables *inside* each region.

Reading order: sort regions, then sort inside a region (enhanced XY-cut, `use_region_detection=True`). Apache-2.0. CPU-possible. A Docling plugin can swap Docling's layout net for PP-DocLayout-V3.
This matches **sibling panels** (left vs right). Token chips *inside* a panel can still flatten.

4.2 End-to-end VLMs that emit structure

They look at the poster instead of sorting OCR lines.

Tool	Size	What it is for	On this laptop
Granite-Docling / SmolDocling	~258M	DocTags with locations; IBM mentions slides/infographics	**Ran.** Splits panels; does not read them. Header hallucination.
dots.mocr	~3B	Layout JSON + reading order; charts/diagrams as SVG	GPU / Tiiny. Not stage 1.

Tool	Size	What it is for	On this laptop
Surya 2	0.65B	Layout labels + reading-order index	Maybe later; **RAIL-M** weights (same licence class v3 flagged for Marker).
olmOCR	~7B	Markdown reading order across columns, figures, insets; brochures in the mix	GPU. Overkill for stage 1.

Comic-panel detectors and HierText-style scene-text models are cousins, not drop-in `extract` Implementations.
Geometric crop (split on the vertical gap, or on Granite/PP-Structure boxes, then OCR each crop) is not a neural product. It is the cheap Fallback that *would* have fixed the JSON merge on this file without a second stack.

5. WHAT "BETTER EXTRACT" HAS TO MEAN FOR THIS CLASS

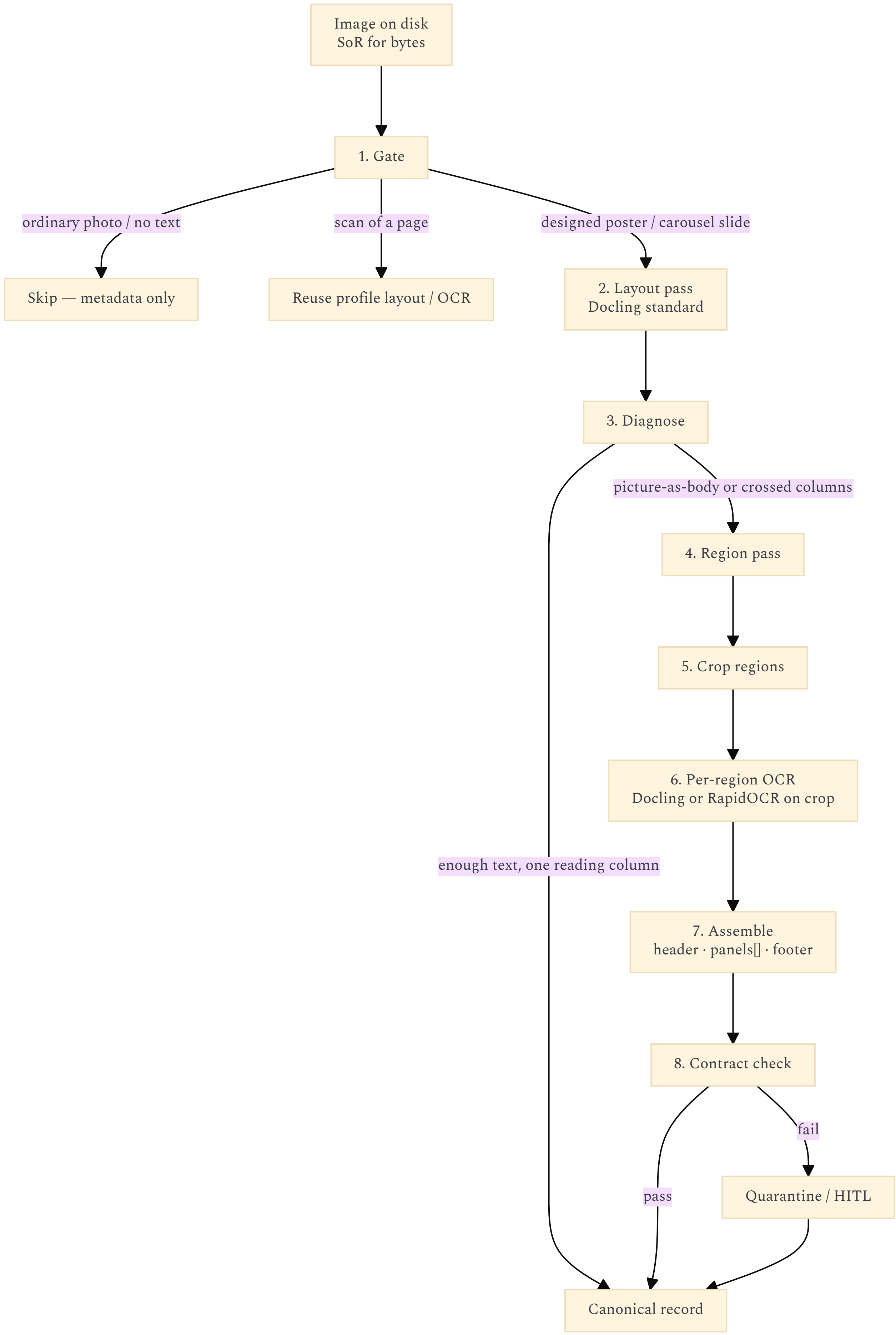
Success is not a higher OmniDocBench score. For a two-panel poster, a gold item passes if:

1. Header paragraph is verbatim (no VLM synonym / contamination).
2. Panels are **separate regions** (not one picture, not interleaved `CONTEXT WINDOW` × 2).
3. Nested labels stay attached to their chip (`PINNED (TOKEN #1)` under the green `The`, `(evicted)` under the empty slot).
4. Twin wells are not merged (`fluent.` must not sit next to `xq7!@#`).
5. Footer punchline is present and ordered last.
6. Locale of the source is kept (fr / en / pt). No English-only store.
7. Original JPEG remains SoR for **bytes**. Extract is a projection with Technology hash, confidence, and a pointer back to the file.
8. Garbage (failed OCR, VLM hallucination) is quarantined (`extract.failed` / KnowledgeGap), not stuffed into Meilisearch.

Markdown export that replaces the diagram with `<!-- image -->` is a **failed extract** for this class, even if JSON secretly has the words.

6. PROPOSED MULTI-STEP DESIGN

One pivot: `extract`. New **profile** (name bikeshed: `poster` / `carousel` / `infographic`). Implementation is a **pipeline of named steps**, each closable. Do not make Granite, Paddle, or MinerU the identity of `extract`.



Each step is an ASC hook-shaped job (`pre_extract/poster`, inner scripts). The UI and the model only see `extract` with `profile=poster`.

Step 1 — Gate (do not OCR the social dump)

Cheap classifier **before** any layout VLM (Gazit: triage is not a 70B). Features that are enough for v1:

- Source tree / sidecar (TODO "document - image - infographic" entity ?).
- Aspect ratio (4:5 / 1:1 slide vs 3:2 photo).

- Optional tiny heuristic: many axis-aligned rectangles (poster chrome) vs few (photo).
- User allowlist: "this folder is educational infographics."

Labels:

Gate	Next profile
No useful text	Skip. Store path + mtime only.
Searchable PDF / EPUB	`plain` / `layout` as today.
Scan of a *page*	`layout` + OCR worker.
Designed slide / carousel / two-panel poster	`poster` (this pipeline).
Scene text in a photograph (sign, meme, UI chrome)	Later: scene-OCR Implementation (PaddleOCR), not MinerU.

Cost cliff: default is **skip**. Indexing is opt-in per tree, as note 17 already required.

Step 2 — Layout pass (Docling, one shot)

Run existing Docling **standard** on CPU. Write:

- markdown (what humans will pack),
- JSON `DoclingDocument` (what we diagnose),
- Technology hash (Docling version + OCR engine + mode).

Do **not** start Granite or PP-Structure yet. Most notes and many simple slides will be good enough here.

Step 3 — Diagnose (the router)

Inspect the JSON, not the markdown.
Signals that the poster pipeline must continue:

- Body is essentially `text + one huge picture` (or `text + picture + caption` with the caption under-ordered).
- Picture children exist but reading order **interleaves** two x-clusters (two `CONTEXT WINDOW` at similar y; two `MODEL OUTPUT`).
- Exported markdown contains `<!-- image -->` while the picture has many text children (export loss).
- Character count of body text ≪ character count of picture children.

If none of those fire: stop. Canonical record = Docling output. This is the Lefèvre karma path (deterministic, cheap).

Step 4 — Region pass (split sibling panels)

Goal: a list of bboxes `{id, role, bbox}` covering **header band, panel_1, panel_2, …, footer band** — not token chips yet.
Implementation order (Fallback chain, YAML `able`):

1. **Geometry** (**always available**). If the diagnose step already sees two x-clusters of picture-children, take the gap. Else: largest vertical gutter in the middle third of the image. No extra model.
2. **Granite-Docling VLM** (**CPU, already in Docling**). Use it **only as a regioner**: keep picture bboxes, **discard its transcribed header**. Proven on 2026-08-30: two panels, empty insides, hallucinated prose.
3. **PP-StructureV3 region detection** (when it earns an eval). Coarse `LayoutRegion`s for newspaper-like siblings. Prefer this over Granite if the gold dozen shows fewer header hallucinations and comparable splits.
4. **MinerU** / **dots.mocr** / **olmOCR** — stage 2, GPU/Tiiny, only if 1–3 fail the gold dozen. Named Technologies, `lan-only` / closable.

Do not run 2+3+4 on every file. One regioner per file, recorded in the extract contract (`regioner=`).

Step 5 — Crop

Crop the original JPEG (lossless-ish PNG crops are fine) per bbox, with a small pad. Files on disk next to the canonical extract:

```
<source>.poster/
  header.png
  panel-01.png
  panel-02.png
  footer.png
  regions.json    # bboxes, regioner, hash
```

Original remains immutable. Crops are derived (Kleppmann).

Step 6 — Per-region OCR

Run **standard Docling or RapidOCR** on **each crop**. A crop of one panel is a single-column mini-page: XY-cut inside it is much less likely to jump to the sibling panel.
Nested boxes *inside* a panel (token chips) may still come out as a line of words. That is acceptable for v1 if:

- left panel text is not mixed with right panel text,
- the fluent sentence is contiguous,
- labels (`PINNED`, `(evicted)`) are at least in the same panel blob.

Optional later: a second, inner region pass on a panel crop if chip-to-label attachment is a measured failure. Do not invent that before the gold dozen.

Step 7 — Assemble the canonical record

Contract (Postgres JSONB + sidecar; files stay SoR):

```
extract_id
source_path
bytes_hash
profile            = poster
locale             = en | fr | pt | mixed
gate               = { label, score }
layout_pass        = { tech, hash }
regioner           = { name, hash, bboxes[] }
ocr_per_region     = [ { region_id, tech, text, confidence } ]

header_text
panels[]           = [ { id, role, text, bbox } ]
footer_text
reading_order      = ["header", "panel-01", "panel-02", "footer"]

warnings[]         = e.g. vlm_header_discarded, markdown_picture_placeholder
confidence
extracted_by       = pivot + Implementation + time
```

Packing / Meilisearch:

- Index **header** + **per-panel text** + **footer** as separate docs or as one doc with `panel_id` filters.
- Do not flatten 400 chip-words into one embedding chunk.
- Keep a pointer to the JPEG; the UI opens the image, not a reconstructed poster.

If Granite (or any VLM) produced a header, **prefer the Step 2 Docling header** when diagnose said the header was already good. Never let a regioner overwrite a better lexical extract of the same band.

Step 8 — Contract check and quarantine

Fail closed (Winteringham / Sanderson):

- Empty panels after region+OCR → `extract.failed`, KnowledgeGap, do not index.
- Header from a VLM that disagrees with Docling's header beyond a small edit distance → keep Docling, log `vlm_header_discarded`.
- Panel text that still contains both `CONTEXT WINDOW` copies in one region → regioner failed; try next Fallback or HITL.
- Locale: Portuguese/French slide forced through English-only OCR config → failed eval (already a revival rule).

HITL is drama: a human can accept a messy panel or mark "diagram, do not index body." The indexer does not mint Claims from carousel OCR (social-media review: a carousel is not a paper).

7. HOW THIS SITS ON THE EXISTING PROFILES

Do **not** overload `figure`. v3 `figure` is "crop + caption + optional VLM," and it explicitly says do not OCR every figure because Docling can. Posters are the opposite: the figure **is** the text.

Profile	When	Engine
`plain`	Digital PDF/EPUB	pdftotext
`layout`	Multi-column papers, office	Docling CPU
`biblio`	Scholarly identity	DOI lookup; GROBID if lookup fails
`table`	Grids that must stay grids	TableFormer / Camelot
`figure`	Photos of pages, charts to *point at*	Crop + caption; OCR gated
`poster`	Designed slides, carousels, two-panel infographics	**This pipeline**
OCR worker	Scans, document-like images	RapidOCR / Paddle; not MinerU-the-product

`index` stays lexical-first (Meilisearch). `poster` writes canonical text files; projections rebuild.

8. STAGE 1 VS LATER (SAME NAMES)

Stage	What may run	What must already be named
1 — this laptop	Gate + Docling standard + diagnose + geometric or Granite **regioner** + crop + RapidOCR/Docling on crops	`extract` `profile=poster`; quarantine; do not index failures
2 — Tiiny / better GPU	PP-StructureV3 if eval wins; dots.mocr / MinerU-VLM / olmOCR as Overflow regioner or inner-panel reader	Same contract. `provider=` / `regioner=` in YAML. Illich: unplug the box, stage 1 still extracts.

Stage	What may run	What must already be named
3 — other contexts	Same pivots, nested allowlists (client corpus vs household)	Do not fork a second ontology for "social OCR."

May wait: running Granite on every gated image (only on diagnose-fire); PP-Structure plugin; any 3B+ VLM; scene-text on memes.
Tracer bullet (Hunt & Thomas): **one JPEG, terminal, this pipeline, compare to §2 table.** The 2026-08-30 file is the first gold item.

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9. GOLD DOZEN (HOW TO FALSIFY THIS NOTE)

Pick ~12 slides from the gated educational set (en + at least one fr or pt). Include: two-panel comparisons, 3+ nested boxes, a dense text-only carousel, a chart-with-caption, a failure (photo with no text).
Score each Implementation as a **composition**, not a vendor:

Composition	Must beat
A. Docling standard only	Baseline (known fail on the attention-sink slide).
B. Docling + diagnose + **geometry crop** + Docling on crops	Should separate panels without a VLM.
C. Docling + **Granite regioner** + Docling on crops	Region split already observed; measure header contamination (must discard Granite text).
D. PP-StructureV3 regioner + crops	Only keep if better than B/C on nested chips *and* headers.
E. MinerU pipeline or VLM on the whole slide	Only keep if it beats D on the dozen *and* still runs as a job.

Kill the composition if it hallucinates the header, merges panels, or needs the 1050 to do 7B inference.

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10. SETTLED VS OPEN

Settled (do not reopen from this slide):

- Docling remains default `layout` for PDFs. Tika is not identity. MinerU is not the social-OCR complement.
- This JPEG class needs a **pipeline**, not a better one-shot markdown dump.
- Granite-Docling: keep as an optional **regioner**, never as the transcriber of the header.
- Glyph OCR (PP-OCRv6 / RapidOCR) is good enough on this example; do not shop OCR engines to fix reading order.
- Original image = SoR for bytes. Extract is a projection. CLI = GUI.
- Do not OCR the whole social dump. Gate.

Open (implementation, not identity):

- Profile name: `poster` vs `carousel` vs `infographic`.
- Whether geometry-first (B) is enough to skip Granite on most files.
- Whether PP-StructureV3 beats Granite as regioner on the gold dozen (eval, then freeze a Technology hash).
- How deep to recurse inside a panel (chip ↔ label attachment).
- Scene-text Implementation for non-poster photos (PaddleOCR vs skipping).
- Whether markdown export should inline panel text or always keep the JPEG as the UI object and only index sidecar text.

Those stay experiments behind `extract` `profile=poster` without changing the IPC vs localhost rule in note 17.